

Summary

AI and LLM engineer with 10 months across four industry internships, shipping both the model and the AI systems that keep it running. Fine-tunes, benchmarks and serves tool-calling LLMs on constrained GPU hardware, and builds the fault-tolerant, observable distributed infrastructure underneath them.

Education

B.Tech, Chemical Engineering | Indian Institute of Technology Madras | 2023 – 2027

Class XII (CBSE) | Hem Sheela Model School | 2023

Class X (CBSE) | Hem Sheela Model School | 2021

Technical Skills

Languages & Tooling	Python, C++, TypeScript, SQL, Bash, Git, pytest, uv
ML & Deep Learning	PyTorch, Transformers, TRL, PEFT, LoRA/QLoRA, Fine Tuning (SFT), Evaluation & Benchmarking
GPU & Training Infra	CUDA, DGX Spark (GB10), Unsloth, FlashAttention-2, Transformer Engine, bitsandbytes
Quantization & Serving	NF4, BF16, FP8, NVFP4, GGUF / Q4_K_M, vLLM, llama.cpp / llama-server
LLM Systems	Tool Calling, RAG & Hybrid Retrieval, GraphRAG, Chonkie, ChromaDB, pgvector, Roboflow
Distributed Systems	Temporal, Ray, Event Sourcing, Airflow, Celery, Exactly Once Semantics
Backend & Web	FastAPI, Django REST, PostgreSQL, Redis, Next.js, React
Infra & Observability	Docker, GCP, Modal, Railway, Playwright, CI/CD, Prometheus, Grafana

Relevant Coursework

LLM Fine Tuning & Optimization • Generative AI with LLMs (DeepLearning.AI) • Machine Learning in Production (MLOps) • AI Agents (Hugging Face) • Mathematics for ML (Linear Algebra) • Deep Learning Specialization (Andrew Ng) • Probability & Statistics for ML (DeepLearning.AI) • Node.js, Express & MongoDB (Backend Bootcamp)

Experience

Otsuka Corporation

Tokyo, Japan • PPO

AI/ML ENGINEER INTERN (SOFTWARE DEVELOPMENT ENGINEER INTERN)

May 2026 – Jul. 2026

- Ran 5 controlled LoRA/QLoRA experiments on Qwen3-8B against **BFCL v4** (NVIDIA **DGX Spark**, GB10), establishing **data composition, not volume** as the central lever for tool-calling SFT. Made strict-format calling native to the model, removing the dedicated format handler the base model needed, at no cost to accuracy: **88.6%** 8-category average, **97.0%** on `simple_python`.
- Shipped **Senpai**, a sales copilot on Otsuka's production SPR schema - a deterministic 7-signal deal-health engine plus a tool-calling LLM over FastAPI and Next.js, served on **Qwen3.6-35B-A3B-NVFP4**; traced repetition-loop degeneration to unset sampling under greedy decode and fixed it by splitting decode policy per round.
- Rearchitected Senpai's serving path off a frozen-context A/B: moved grounded synthesis to a **Q4 8B (2.72× wall, 3.8× decode, at grounding parity)** while keeping tool selection on the 27B, which the 8B could not do reliably - attribution showed retrieval and tool execution were under **1%** of turn latency, so the two generations per turn were the real bottleneck.
- Root-caused catastrophic **FP8 gradient explosion** on GB10 (`sm_121`) after eliminating **8 hypotheses**: MXFP8 is unimplemented for the arch, so Accelerate silently fell back to `DelayedScaling`, whose `amax_history` gradient checkpointing double-updates on recompute. Moved production to Unsloth NF4 QLoRA (**22% faster, 45 GiB lighter** than HF+PEFT).

Gravton Labs

Ontario, Canada (Remote)

AI ENGINEER INTERN

Feb. 2026 – May. 2026

- Primary engineer on the **citation attribution engine** behind a live AI-search-visibility product: normalized extraction across **5 LLM provider families** (OpenAI, Anthropic, Gemini, Perplexity, xAI) plus Google AI Overviews behind one registry, gated on retrieved grounding so ungrounded URLs never score, with span-level support attribution and automatic competitor discovery from unrecognized cited domains.
- Sole author of the Reddit, Quora and YouTube intelligence pipelines - three Django apps behind their own Airflow DAGs (YouTube alone is **17 tasks**) - with deterministic 5-signal community-authority scoring and entity-mention answer units exposed downstream as a read-only contract.
- Built and owned the **production crawler**: adaptive page discovery (DOM, sitemap, Apify fallback) feeding a heuristic best-first scorer, so a fixed page budget is spent on commercial-intent pages instead of blog sprawl - over a three-tier pooled **Playwright** runtime (**2 browsers / 100 contexts / 250 pages**), later cut **5–10×** after auditing pool exhaustion under load.
- Drove the **Airflow** migration and took the platform to production: containerized the Django, Celery and Airflow stack, moved crawl, enrich and citation stages onto scheduled DAGs with idempotent persistence, then hardened the deploy (Gunicorn, locked-down Traefik, internal services off the host network).
- Ported the crawler and insights services from Encore TypeScript to FastAPI on Railway over Supabase, dispatching heavy stages to **Modal** serverless; owned customer onboarding and closed a cross-brand-set data leak that surfaced out-of-scope competitors in customer results. Both platforms run **in production today**.

OctonData

San Francisco, CA (Remote)

SOFTWARE ENGINEER INTERN

Oct. 2025 – Dec. 2025

- Primary engineer on a document intelligence platform on **GCP** (Cloud Run, Cloud SQL, Cloud Build CI/CD); shipped the Tax CPA Parser to **10+ U.S. CPA firms**, cutting manual review time **40%+**.

- Extended **od-parse**, the in-house parser, from PDF and image input to Excel, DOCX, PPTX and CAD vector formats behind a classifying router with vector-vs-raster PDF triage; shipped it as a Dockerized FastAPI service.
- Owned od-parse's **mechanical-drawing pipeline** - Roboflow annotation detection, batched Gemini multimodal verification, and a full-image rescan recovering annotations the detector missed - batching all patch verification into one call to stay under provider rate limits.
- Grounded a **CrewAI** multi-agent tax analysis system (return analyzer, IRS code researcher, strategy analyst) on Gemini over a semantic chunker (Chonkie plus Gemini embeddings, recursive fallback), behind JWT role-based access and organization-level multi-tenant isolation.

Tecnod8.ai

IIT Mandi Incubated (Remote)

MACHINE LEARNING INTERN

Sep. 2025 – Oct. 2025

- Built multilingual document parsing across **5+ languages** with YOLOv10, PP-DocLayout-L and PaddleOCR - RTL and Devanagari support, ensemble layout inference with rotation handling and adaptive scaling.
- Integrated **Qwen2.5-VL** for table, figure and chart extraction, embedding structured outputs via Gemma into ChromaDB for retrieval.
- Trained the detector to bounding-box document structure as distinct classes - **titles, section headers**, paragraphs, tables and figures - and raised its accuracy through dataset auditing, **active-learning** reannotation of low-confidence regions, and per-class error analysis.

Projects

Ancora – Fault-Tolerant Runtime for Durable AI Workflows

Open Source

PYTHON · TEMPORAL · RAY · FASTAPI · OPENTELEMETRY · NEXT.JS

Jul. 2026 – Present

- Built a durable-execution runtime for AI workflows on **Temporal** (event-sourced) and **Ray** (distributed compute), exposing retries, checkpointing and resumption through one decorator-level API - **5 services, 284 passing tests**, `mypy --strict` and `ruff` clean.
- Shipped a chaos engine that **machine-verifies durability** rather than asserting it: SIGKILLS live worker containers mid-run, then checks *no-lost-state*, *no-re-execution* and *exactly-once effects* straight from Temporal history against an SLO. Cut failover detection **9.5×** (60 s to 6 s) via activity heartbeating.
- Made a killed activity resume **mid-step** rather than restart by carrying its checkpoint in the Temporal heartbeat; added durable human-approval gates and replay tests pinning workflow determinism.

TinyServe – From-Scratch LLM Inference Runtime on llama.cpp

Open Source

PYTHON · LLAMA.CPP · FASTAPI · PROMETHEUS · OPENTELEMETRY

Aug. 2026 – Present

- Built continuous batching, block-based KV-cache accounting and pluggable fair scheduling (**FIFO / Priority / WFQ**) from scratch over llama.cpp, validating the riskiest bet - real multi-sequence batched decode - with a proof-of-concept before building the scheduler around it; **WFQ bounds unfairness 2.54×** against strict Priority's **3.87×**.
- Profiled with `py-spy` and cross-checked against a self-emitted Prometheus metric: **97.2%** of wall time inside `llama_decode()`, confirming the scheduling layer wasn't the bottleneck; admission control rejects exactly at the KV-budget boundary under burst load (**42/64 admitted**).

Interactive 3D Portfolio – Real-Time WebGL Workspace

Personal Project

NEXT.JS 15 · REACT 19 · THREE.JS · WEBGL

Jul. 2026 – Present

- Profiled frame spikes across a 200-step orbit, attributing cost per feature; no single source dominated, so the fix had to be cumulative - **p99 28.4 to 21.2 ms**, zero dropped frames.
- Caught a fixed DPR ceiling rendering *below* native on $2\times$ panels; replaced it with a `devicePixelRatio`-relative target holding constant $1.3\times$ supersample across five display shapes, behind a Draco and WebP pipeline (**25 MB to 4 MB** deployed).

R.E.A.C.H. – AI-Powered Emergency Response Platform

IITM Nirmaan Cohort

CO-FOUNDER & AI LEAD

Apr. 2025 – Present

- Fine-tuned **Whisper** on noisy emergency call audio for real-time transcription; integrated BART summarization with call spoof detection at **78% precision**.
- Built the real-time SOS dispatch and live responder-tracking backend on **Socket.IO**, Leaflet/OpenStreetMap and the **OSRM** routing engine - computing driving routes and streaming interpolated position updates over WebSockets; added a metadata-based fake-report filter validating EXIF GPS/timestamp against claimed incident locations. Selected from **200+ startups** by IITM NIRMAAN; drove MVP delivery across a 5-member team.

Extracurricular Activity

Sourced 200+ alumni founders via Tracxn for batch internship leads • Mentored 10+ students via IITM Shiksha Prayas • Edited 10+ videos for IITM TV, reaching 5,000+ total views